KENYA MEDICAL TRAINING COLLEGE
FACULTY OF REHABILITATIVE SCIENCES
DEPARTMENT OF MEDICAL ENGINEERING

FINAL QUALIFYING EXAMINATION
FOR
CERTIFICATE IN MEDICAL ENGINEERING TECHNOLOGY

PAPER: CLINICAL ENGINEERING THEORY

DATE: 16th June 2022 **TIME: 3 HOURS (9:00AM – 12:00 PM)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. Write the **EXAMINATION NUMBER** given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

Clinical Engineering

SECTION 1: MULTIPLE CHOICE QUESTIONS (40 MARKS)

1. The following are functions of water in a dental chair, which one is not?
   a) Cooling at the teeth during grinding
   b) Washing away loose particles from the treatment area
   c) Flushing the cuspidor bowl
   d) Operate the pneumatic controls of instruments

2. The filter on top of the calibrated mercury reservoir is used to?
   a) Ensure that mercury cannot pour out or enter the cuff
   b) Ensure mercury remains stable
   c) Ensure the mercury level does not drop
   d) Ensure the air bulb is operated easily

3. Why do split phase single phase induction motors not commonly used in the medical suction machine? because:-
   a) Has a high speed
   b) Its speed cannot be controlled
   c) Produces sparks
   d) Produces noise while running

4. When measuring pressure, all instruments measure
   a) Under Pressure
   b) Over Pressure
   c) Absolute Pressure
   d) Low Pressure

5. Which one is the correct maintenance done on a vane pump used in medical suction machine
   a) Replenish the oil and clean the vanes
   b) Clean the piston head
   c) Replace worn out diaphragm
   d) Replace the carbon brushes

6. What is the symptom that indicates there is a leakage outlet valve in Bp machine?
   a) Mercury column drops slowly without opening the release valve
   b) Mercury column drops fast; bigger leakage does not even allow a pressure creation in the cuff
   c) Mercury column drops fast sometimes no pressure can be created
   d) No, or only little pressure can be created

7. What is the purpose of using heat sensitive strips during autoclaving?
   a) To check on quality of the products
   b) To check correct pressures of the machine
   c) To check correct temperature of the machine
   d) To check correct exposure time

8. What type of balance is used for measuring foodstuffs?

   a) Spring balance
   b) Lever balance
   c) Beam balance
   d) None of the above

9. The following precautions put in place to avoid miss-identification of medical gases which one is not?

   a) Pin index
   b) Non-inter changeable terminals
   c) Color code
   d) Connecting cylinders in a manifold

10. What causes mercury in a sphygnomanometer to bubble when pumped to rear maximum?

   a) Leaking column filter
   b) Little mercury in the reservoir
   c) Too much mercury in the reservoir
   d) Faulty locking device

11. Which type of electrical connection does three phase heaters used in the autoclaves connected?

   a) Delta
   b) Star
   c) Star-delta
   d) All of the above

12. The most widely used effective, economical and reliable method of sterilization used in the health care setting is

   a) Gas plasma
   b) Ethylene oxide
   c) Parasitic acid
   d) Steam

13. Why is it important to earth medical equipment?

   a) Protect the user for electric shock
   b) Protect the equipment in case of earth leakage
   c) Act as a return in case of earth leakage
   d) All of the above

14. Which of the following pumps require least maintenance attention?

   a) Diaphragm
   b) Pistol pump
   c) Rotary pump
   d) Venture pump

15. Which one is the best confirmation test for electric heater in a good working condition
    a) Check conformity between two terminals
    b) Check conformity between terminal and the body
    c) Connect to electricity
    d) Both A and B

16. The following are supplies to dental department except?
    a) Vacuum
    b) Water
    c) Oxygen
    d) Compressed air

17. At what range will you multimeter to check if the Rheostat of a laryngoscope is working
    a) Ampere range
    b) Voltage range
    c) Ohms range
    d) Frequency range

18. Vacuum is one of the supplies in a medical department. Its pipeline color code is
    a) Blue
    b) Yellow
    c) Orange
    d) Grey

19. Knite biomedical students carried out the following tests on a transformer of a microscope and got the following results
    i. No continuity in the secondary coil
    ii. Has continuity in the primary coil
    iii. Has continuity between the coil and the body

Which one of the above best test to show that transformer is faulty
    a) i only
    b) i and ii only
    c) i and iii
    d) Ii and iii

20. One of the following is NOT task of the compressed air supplied to the dental unit. Which one is it?
    a) Action as air cushion for some turbine motors
    b) Combining with water to effect a mist spray for cleaning of the treatment area
    c) Flushing the cuspidor bone
    d) Operating the pneumatic controls and instruments which may be part of the dental unit

21. What is the purpose of bottle selection in the machine suction machine?
    a) To control the flow of the fluid to the bottle
    b) Exchange the bottle once the other one is full
    c) It is anti-over flow device
    d) Allows the fluid to be discharged for the bottle

22. The following statements are about electro-cardiograph (ECG) Which one is false?
    a) It records the electrical activities of the heart
    b) It provides valuable information about cardiac disorder
    c) It measures voltage difference in millivolts
    d) It records the amount of oxygen in the blood

23. Which one of the following is commonly used to control the temperature in an electric infant incubator?
    a) Tachogenerator
    b) Thermistor
    c) Rheostant
    d) Fan failure device

24. The following are laboratory equipment. Which one is WRONGLY matched with its function?
    a) Spectrometer- used in the laboratory to determine the concentration of a substance in a solution
    b) Autoclave – used for sterilization
    c) Dry air oven-used in sterilizing glass and metal containers
    d) Bacterial incubator- killing of microbes on instruments

25. Which one of the following is not supply system dental department in the hospital?
    a) Vacuum
    b) Medical air
    c) Compressed air
    d) Drainage

26. One of the following is not an alarm indicator used in infant incubators. Which one is not?
    a) Fan failure alarm
    b) Power failure alarm
    c) Incubation period over alarm
    d) Overheat alarm

27. For what purpose diathermy is used?
    a) Therapeutic and diagnostic
    b) Surgical and therapeutic
    c) Diagnostic and surgical
    d) Diagnostic and rehabilitative

28. Which of the following faults would be the cause of endogenous burns in electro-surgery?

a) Patient plate too small with respect to Hf power
b) The patient plate not covering the patient tissues with its entire area
c) The patient plate is in poor condition
d) All of the above

29. Which of the following parts do not belong to the anaesthesia machine?

a) Fail safe valve
b) APL
c) Flow sensor
d) Active electrode

30. It is advisable when installing a microscope, a study level table is recommended so as to

a) Not block air vents on the underside of the bare of microscope
b) Obtain clear vision of specimen
c) Avoid damage of specimen
d) Control operation in case of emergency

31. Which one of the following statement is false about single chamber autoclave?

a) End products are wet
b) End products are dry
c) Sterilizing and steam generating chamber is same
d) Leads to be sterilized is placed in steam sterilization chamber

32. The oxygen concentration of an infant incubator is

a) 75%
b) 40%
c) 80%
d) 36%

33. What is the function of cuspidor bowl in the dental unit?

a) To drain water used for rinsing patient mouth
b) Contains pre-heated water to allow patient rinse their mouth
c) To remove unwanted fluids as particles from the mouth
d) Both A and B

34. Which of the following time-temperature relationship for steam sterilization is correctly marched?

a) 121° C for 10 min at 1 bar
b) 126°C for 15 min at 10psi
c) 134°C for 3 min at 3 bar
d) 160°C for 10min at 1 bar

35. The following are examples of unipolar leads, except
   a) AV_R
   b) AV_L
   c) AV_F
   d) AV_D

36. The following are the basic parameters involved in the effective measurement of PH, which one is not
   a) Actual molar concentration of hydrogen ions
   b) Actual molar concentration of hydroxyl ions
   c) The disillusion constant of the acid
   d) Temperature

37. The following are statements about invasive blood pressure measurements. Which one is not true
   a) It requires surgery
   b) It is painful
   c) It entails continuous monitoring
   d) It does not require expertise

38. Which among the following is a method of manufacturing carbon dioxide (CO_2) gas?
   a) Fractional distillation of air
   b) Thermal decomposition of ammonium nitrate at 250°c
   c) As a combustion gas from burning fuel
   d) Chemical decomposition of oxygen compounds

39. Surgery hand piece of a dental chair is supplied by
   a) Electricity
   b) Compressed air
   c) Water
   d) Vacuum

40. Which part of BP machine transmits the pressure to the upper arm of a patient?
   a) Air bulb
   b) Bandage cuff
   c) Inlet and outlet valve
   d) Manometer

SECTION 2: SHORT ANSWER QUESTIONS (40 MARKS)

41 Digital blood pressure monitor should be charged to ensure that it works properly and records accurate readings. During the normal rounds in the outpatient department in your facility, you noted that the blood pressure monitor is always charged most of the time. The nurse on duty explains to you that they attend to so many patients and thus the blood pressure monitor should never run out of charge. What advice will you give the nurse on when she should charge it and stop charging the medical equipment? 5 marks

Note

i) There are colours that indicate when to charge and when not to charge.

ii) The blood pressure monitor should be charged when the charge drops to 25%.

Give your advice based on the colour.

42 A nurse in the new born unit complains that weighing equipment gives incorrect weights of the infants whose weights are monitored. What are the probable causes? Give a solution to every cause 5 marks

43 You are required to install electronic weighing balance in the hospital laboratory department where you work as a biomedical technician. For the satisfactory installation and use of the balance, what are some of the requirements that you should consider? 5 marks

44 What is the probable cause to an autoclave which is not fully pressurizing? 5 marks

45 State the specific features of a patient's dental chair 5 marks

46 Some parts and components of the dental unit are degraded or deteriorated depending on the frequency of use. Annual check-up and maintenance, as well as replacement of consumable parts are required. Name this parts 5 marks

47 a State the reason why high frequency current is used in surgical diathermy 2 marks

b What are the three factors that contribute to the fluctuation of output voltages of surgical diathermy? 3 marks

48 If low pressure in the anesthesia machine is detected, which parts should be examined? 5 marks

SECTION 3: LONG ANSWER QUESTION (20 MARKS)

49 a The measuring or detector electrode in a pH meter requires periodic maintenance of the conducting solution to obtain precise readings. Outline the steps for replacing the electrolyte solution — 12 marks

b Outline the TWO most commonly used types of mechanical balances in the laboratory department — 6 marks

c The machine circuit of anaesthesia machine can be broken down into two parts, name them — 2 marks